

TABLE OF CONTENTS

TABLE OF CONTENTS 1

1- LASER DIODE LIGHT ALIGNMENT PROCEDURES 2

 1-1 Tools Required 3

 1-2 Procedure..... 3

REVISION HISTORY 11

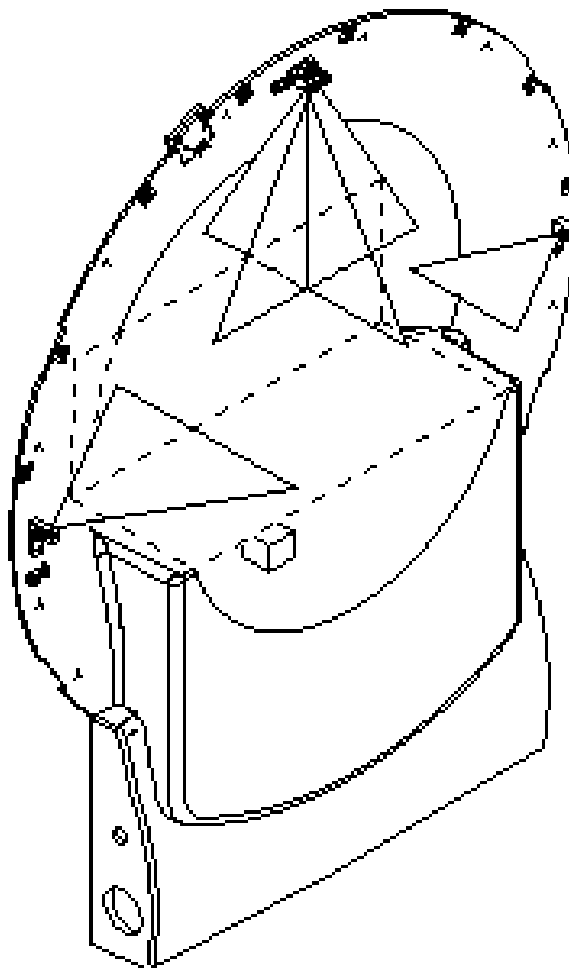


The Laser Diode Lights are factory pre-set. Refer to procedure for Patient Alignment Checks for CX magnet first. If functional checks are good do not perform alignment procedure.

Description - Procedures for aligning the patient alignment lights and checking the power supply voltage. For Non-CX systems refer to procedure for Patient Alignment Lights for Non-CX Magnet.

1- LASER DIODE LIGHT ALIGNMENT PROCEDURES

The Laser Diode Lights are mounted on the front magnet frame as shown in Illustration L2548B. All three lights are identical.



LASER ALIGNMENT DETAIL

LASER ALIGNMENT DETAIL
ILLUSTRATION L2548B

1-1 Tools Required

- SPT Short Loader Cylinder, 2135652

Note

The SPT Short Loader Cylinder is part of the SPT Short Loader Assembly (2125244). This hardware, though shipped with Signa systems, is the property of GE. It is available for customer purchase.

- Nonmagnetic scale (48 in. nominal)
- Nonmagnetic 12 in. ruler
- Nonmagnetic screwdriver
- Nonmagnetic 0.050 in. Allen wrench
- Nonmagnetic open-end wrench
- Multimeter
- Protective eyewear
- Nonabrasive gloves
- Nonmagnetic straight edge (min. 43 in.; max. 45 in.)

WARNING!

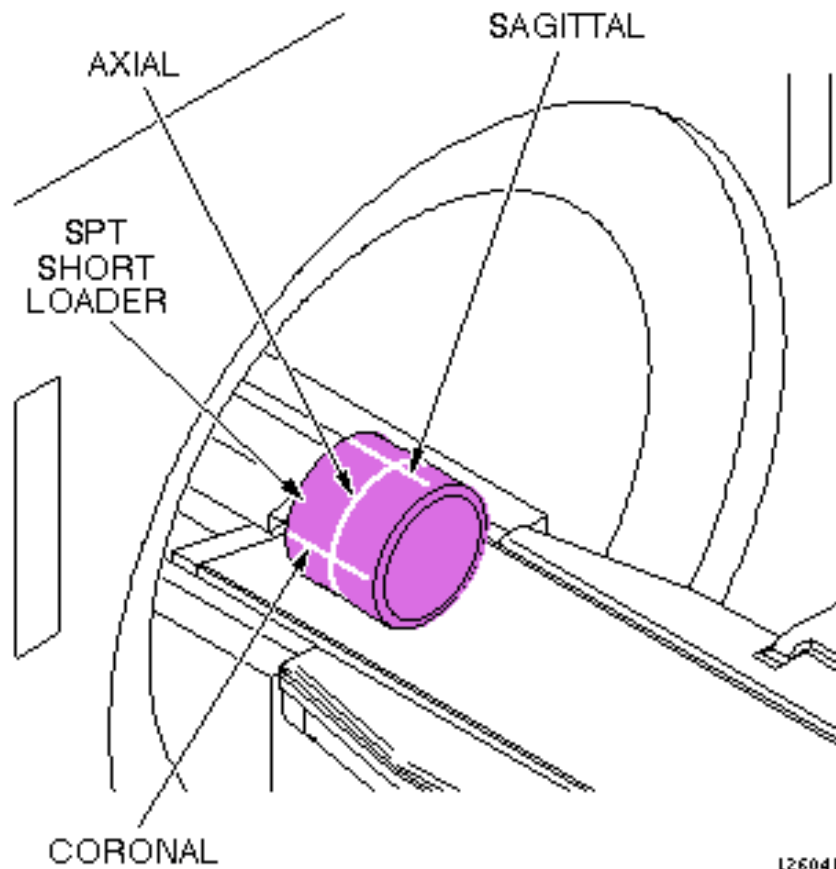
POISON HAZARD! THE PHANTOM CONTAINS NICKEL, A SUSPECTED CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

1-2 Procedure

1. Open Magnet Enclosure front cover.
2. Turn on power to the Laser Light Assemblies High Current Power Supply Board (MR1A14A2) by switching on CB6 on Magnet Enclosure Power Supply Module (MR1A14) at the RF/Pen Cabinet.

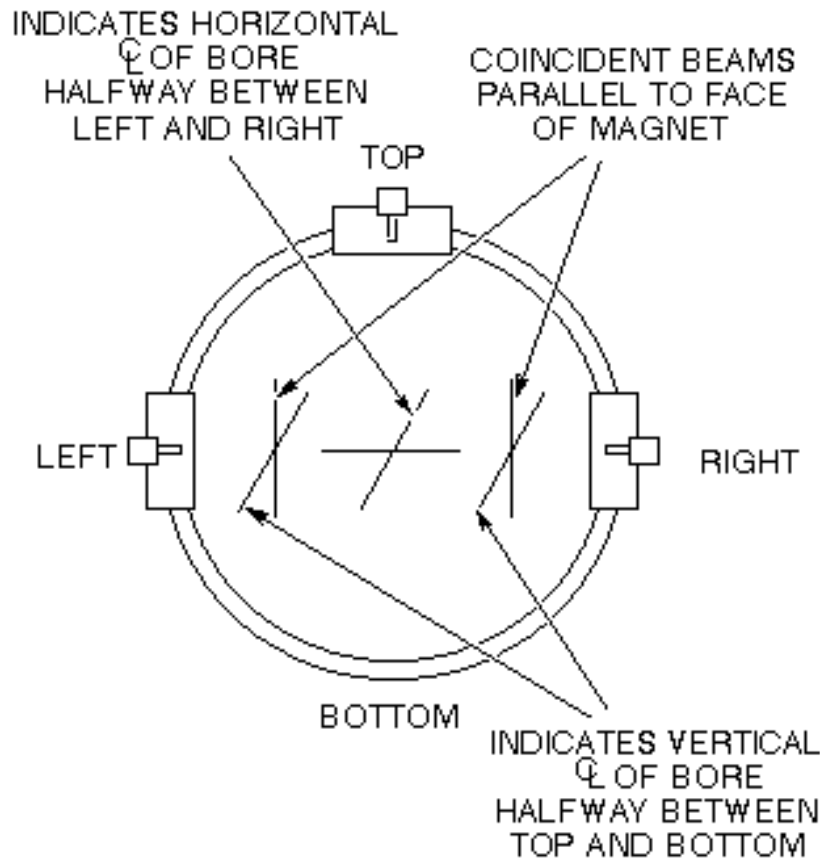
For RF/Pen II and RF/PDU Cabinets: Switch on circuit breaker MR1 A20 CB4 at rear of RF/Pen II and RF/PDU Cabinets (SSM).

3. Place SPT Short Loader Cylinder on patient table. See Illustration L2604B.



LASER WRAPAROUND BEAM PATTERN
ILLUSTRATION 2604B

4. Press ALIGN ON pushbutton on front of magnet enclosure.
5. Close Magnet Enclosure Front Cover and observe light beam at top aperture. Verify that light beam is unobstructed, and that there is no interference with the magnet cover. See Illustration L2603A and Illustration L2604B for the desired beam pattern at the front of the Magnet Enclosure.



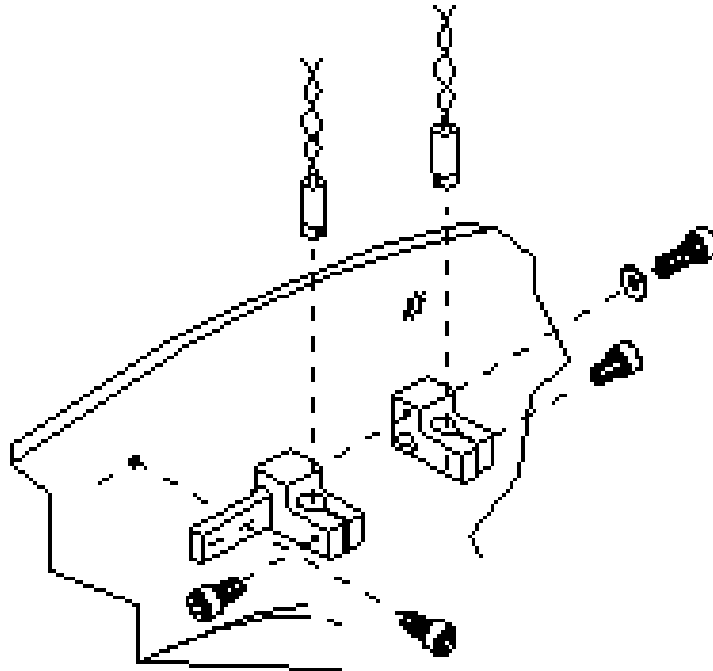
L2603A

LASER LIGHT BEAM PATTERN
ILLUSTRATION L2603A

6. Adjust the three assemblies in the lateral direction so that the beams are centered in the magnet bore as shown in Illustration L2603A and Illustration L2604B above. Illustration L2604B shows the wraparound pattern with the three laser beams perfectly aligned. The top laser sagittal beam should intersect the bore at the horizontal centerline. This can be measured by placing a nonmagnetic scale horizontally across the face of the bore and measuring the intersection of the beam.

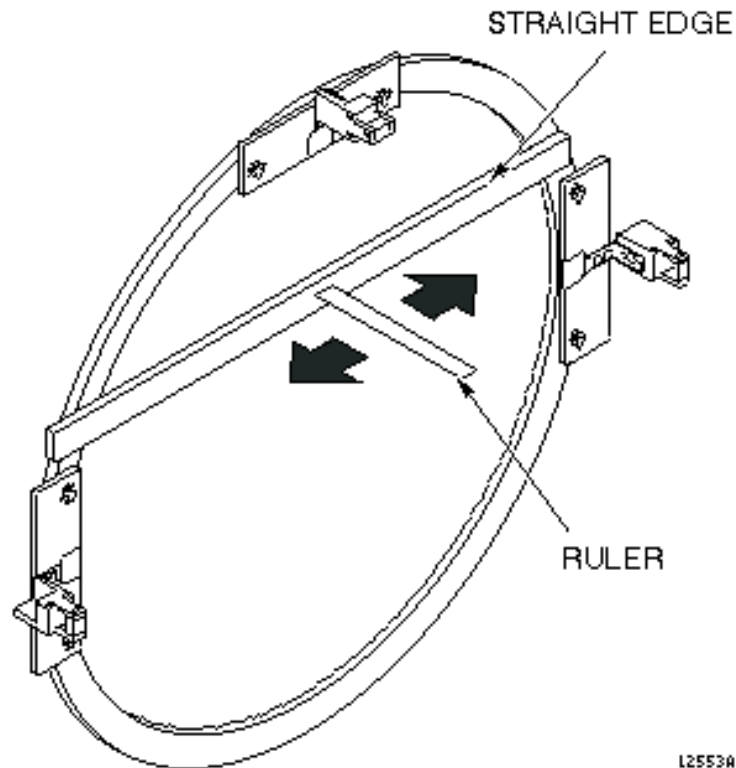
The left and right laser coronal beams should intersect the bore at the vertical centerline of the bore opening. A non-magnetic scale placed vertically across the face of the bore will determine the alignment of these beams to the vertical centerline of the bore.

7. Tighten alignment screws when adjustment is complete. Apply Loctite to screws before tightening. Do not turn or clamp on laser light lens caps or damage will occur. Pry open bracket with screw driver, then turn laser diode. See Illustration L2604C.



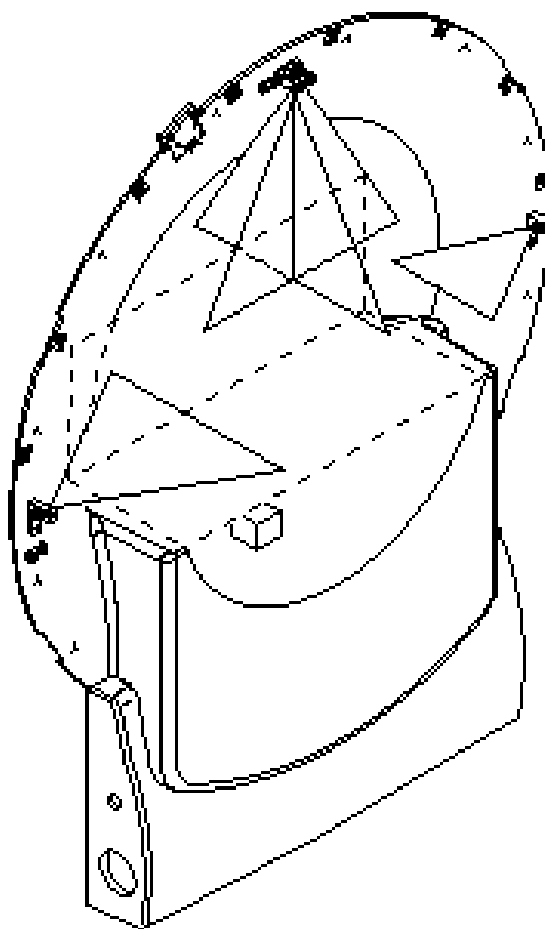
LASER DIODE ASSEMBLY DETAIL
ILLUSTRATION L2604C

8. Using a nonmagnetic straight edge, measure axial beam squareness to face of magnet as follows:
 - a. Place straight edge across the face of the magnet frame, as shown in Illustration L2553A.



UPPER LASER ASSEMBLY ALIGNMENT
ILLUSTRATION L2553A

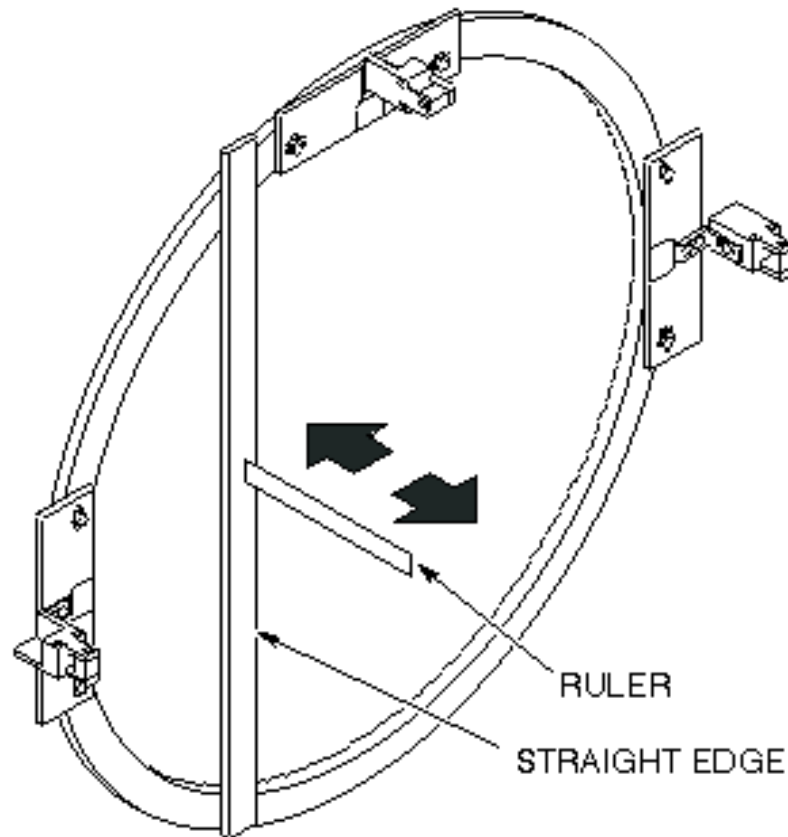
- b. Place a nonmagnetic ruler against the straight edge, and check the squareness of the top laser axial beam by sliding the ruler along the length of the straight edge. Take several readings to ensure that axial beam strikes ruler at exact location at each measurement point; this indicates that beam is square to face of magnet.
- c. If beam alignment is off, loosen rear set screw and gently move laser light until beam is aligned. (See Illustration L2548B.)



LASER ALIGNMENT DETAIL

LASER ALIGNMENT DETAIL
ILLUSTRATION L2548B

- d. Tighten rear set screw.
- e. Repeat this procedure for left and right Laser Light Assemblies, by placing the straight edge against the magnet face as shown in Illustration L2554A. Hold straight edge in one hand and, in the other hand, slide a ruler up and down the face of the straight edge, observing that the axial beam strikes the ruler in the exact location at each measurement point, to ensure that beam is square to face of magnet.



L/R LASER ASSEMBLY ALIGNMENT
ILLUSTRATION L2554A

- f. If measurement points are off, loosen rear set screw and gently move rear lens alignment pin until beam is aligned.
 - g. Tighten rear set screw on left and right Laser Light Assemblies.
 - h. Perform final alignment by loosening longitudinal alignment screws on bracket of left and right Laser Light Assemblies to allow free in/out movement (see Illustration L2548B above). Align left and right axial beams with the top axial beam. Adjust assemblies to ensure that beam path is normal (see Illustration L2603A and Illustration L2604B above).
 - i. Tighten bracket screws when beam paths are normal.
9. Perform a final check of beam pattern, as shown in Illustration L2603A and Illustration L2604B above. Further adjustments may be required to bring beam patterns into exact alignment by repeating entire procedure.
 10. Remove all tools and close Magnet Enclosure Front Cover.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 16, 1998	J. Saperstein	Initial conversion from Toolbook to Word.